PHD125 Pharmaceutical Organic Chemistry · Introduction to Organic Molecule

Before class: carbon, the builder

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

1 · Read first (5 min)

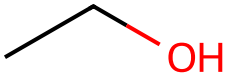
Carbon makes four bonds

Carbon has **four valence electrons**, so it shares electrons to form **four covalent bonds**. It bonds strongly to other carbon atoms, so it builds chains, branches and rings, and it also bonds to H, O, N, S and the halogens.

Compounds built on carbon are called **organic compounds**. There are millions of them, from methane (CH_4) to DNA.

Chemists draw molecules in shorthand

Ethanol can be written as C_2H_6O, as CH_3CH_2OH, or as a zig-zag line ending in OH. In the zig-zag (line) drawing, every corner and end is a carbon and the hydrogens on carbon are hidden. You will learn to read all of these in the lecture.



Why this matters in pharmacy

Almost every medicine you will dispense is an organic compound: paracetamol, aspirin, antibiotics, vitamins. Small groups of atoms in these molecules, called **functional groups**, decide whether a drug dissolves in water, how it is absorbed and how long it lasts on the shelf.

2 · Readiness check

Ten quick questions: PHD115 basics you need today, and a preview.

Q1 Recap

How many valence electrons does a carbon atom (Group 4A) have?

A. 2
B. 4
C. 6
D. 8

Q2 Recap

A covalent bond is:

A. a transfer of electrons from a metal to a nonmetal
B. a shared pair of electrons between two atoms
C. the attraction between two ions
D. a bond between two metal atoms

Q3 Recap

How many covalent bonds does each atom usually form?

Options: 1 · 2 · 3 · 4

H _____

O _____

N _____

C _____

Q4 Recap

A C–O bond is polar because oxygen is more electronegative than carbon.

True / False

Q5 Recap

Which property is typical of an ionic compound such as NaCl?

A. Low melting point
B. Burns easily
C. High melting point
D. Insoluble in water

Q6 CH1-LO1

Organic chemistry is now defined as the chemistry of compounds that contain:

A. oxygen
B. carbon
C. nitrogen
D. living cells

Q7 CH1-LO1

Carbon dioxide (CO_2) is classed as an organic compound.

True / False

Q8 CH1-LO2

Write the molecular formula of CH_3CH_2OH. Type it on one line, e.g. C2H6 for C_2H_6.

_____

Q9 CH1-LO3

Which is the functional group of an alcohol?

A. –OH

B. –COOH

C. $-NH_2$

D. C=C

Q10 CH1-LO5

Name one medicine or everyday product you think is made of organic compounds, and say why.

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Adapted from LibreTexts *Introduction to Organic Chemistry – Alkanes* 12.1, 12.2 and 12.4 (Sharpe Elles, from Ball, Hill & Scott) and *The Basics of GOB Chemistry* (Ball, Hill & Scott) 12.1, 12.4 and exercises 12.E, CC BY-NC-SA 3.0. Pharmacy items are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 Recap

How many valence electrons does a carbon atom (Group 4A) have?

A. 2

B. 4

C. 6

D. 8

Group 4A → 4 valence electrons.

Q2 Recap

A covalent bond is:

A. a transfer of electrons from a metal to a nonmetal

B. a shared pair of electrons between two atoms

C. the attraction between two ions

D. a bond between two metal atoms

Covalent bonds form when atoms share electrons.

Q3 Recap

How many covalent bonds does each atom usually form?

H → 1; O → 2; N → 3; C → 4

H 1, O 2, N 3, C 4: each reaches a duet or an octet.

Q4 Recap

A C–O bond is polar because oxygen is more electronegative than carbon.

True / False → **True**

Oxygen pulls the shared electrons towards itself, giving O a partial negative charge.

Q5 Recap

Which property is typical of an ionic compound such as NaCl?

A. Low melting point

B. Burns easily

C. High melting point

D. Insoluble in water

Strong attractions between ions give ionic compounds high melting points (NaCl melts at 801 °C).

Q6 CH1-LO1

Organic chemistry is now defined as the chemistry of compounds that contain:

A. oxygen

B. carbon

C. nitrogen

D. living cells

Today 'organic' means carbon-containing, whatever the source.

Q7 CH1-LO1

Carbon dioxide (CO_2) is classed as an organic compound.

True / False → **False**

CO_2, CO, carbonates and bicarbonates are treated as inorganic, by convention.

Q8 CH1-LO2

Write the molecular formula of CH_3CH_2OH. Type it on one line, e.g. C2H6 for C_2H_6.

C_2H_6O

2 C, 6 H, 1 O. (C_2H_5OH is also accepted.)

Q9 CH1-LO3

Which is the functional group of an alcohol?

A. –OH

B. –COOH

C. $–NH_2$

D. C=C

Alcohols contain an –OH group on a carbon chain.

Q10 CH1-LO5

Name one medicine or everyday product you think is made of organic compounds, and say why.

Any carbon-based product is right: paracetamol, aspirin, vitamin C, soap, cooking oil, plastic bottles. They are built on carbon skeletons.

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